Trial SET #1 (math) :



- Don't use calculator and any other materials, including formulae and tables;
- You have only 60 minutes to complete the test;
- Try to do your best and Good Luck!!!

You should be familiar with:

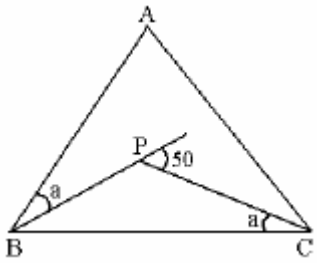
- word problems;
- angles in triangles;
- number properties;
- combinatorics;
- integrals;
- trigonometry;
- arithmetic sequences;
- roots;
- areas in quadrilaterals;
- circles;
- logaritms;
- derivatives;

Name of the student _____

1. A pharmacist wants to prepare 100ml of a 10% acid solution. How much of a 5% solution and a 15% solution should be used to form the desired mixture?

A. 60, 40; B. 45, 55; C. 50, 50; D. 80, 20; E. 40, 60;

2. In the given triangle ABC, $\angle P = 50^0$, $\angle ABP = \angle PCB = a$, find $\angle B$



A. 50; B. 60; C. 70; D. 100; E. 130;

3. Find the value of $28^2 - 72^2 + 995$.

A. -3500; B. -3405; C. 1262; D. -3228; E. 1304;

4. There are four roads from Droben to Northwilk and five roads from Northwilk to Millon. If Kamila drives from Droben to Millon and back, passes through Northwilk in both directions, how many different routes are possible for the round trip?

A. 16; B. 32; C. 160; D. 400; E. 560;

5. How many numbers more than 200 and less than the largest three-digit number are divisible by 7?

A. 113; B. 114; C. 112; D. 115; E. 120;

6. Find the correct equation.

1. $2\sin^2 x + \cos^2 x = 2 - \cos^2 x$. 2. $\mathrm{tg}x + \mathrm{ctg}x = 1$.

3. $\sqrt{\frac{1-\cos x}{1+\cos x}} = 2\mathrm{tg}x$. 4. $\sin^2 x \cdot \cos^2 x = 1$.

A) 4. B) 1. C) 3. D) 2. E) none of them

7. Let the operation $\Uparrow$ be defined by $a \Uparrow b = \frac{a-b}{a+2b}$ for all numbers a and b, where $a \neq b$. If $3 \Uparrow 2 = x \Uparrow 1$, what is the value of x?

(A) 1.5
(B) -2.5
(C) 1
(D) 0
(E) -1.5

8. Identity $(a+b-c)^3 = a^3 + b^3 - c^3 + 3(a+b-c)(ab-ac-bc) + Habc$ is true for all values of *a, b* and *c*. Find the value of *H*.

A. 1; B. -1; C. 3; D. 6; E. -6;

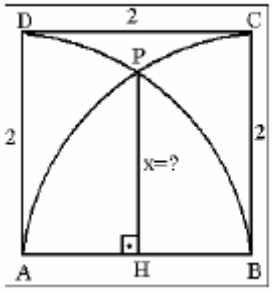
9. A right circular cylinder with a radius of 1 and a height of 1 has a volume that is most nearly the same as the volume of a rectangular solid with dimensions

A. 1 by 1 by 1 B. 1 by 1 by 2 C. 1 by 1 by 3
D. 1 by 2 by 2 E. 1 by 2 by 3

10. What is the remainder when n^3+n^2+n+1 divided by n^2+n+1?

A. n + 1; B. n - 1; C. 3n + 2; D. 2; E. 1;

11. ABCD is a square with side 2 cm, A and B are center of the quarter circles shown in figure. If $PH \perp AB$ then find PH.



A. $\sqrt{2}$; B. $\frac{\sqrt{3}}{2}$; C. $\sqrt{3}$; D. $2\sqrt{3}$; E. $\sqrt{5}$;

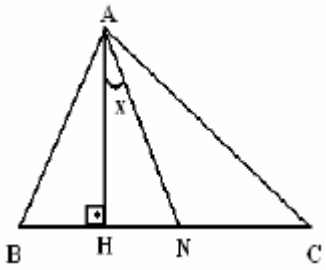
12. The circumference of a circle is 72 cm and the length of an arc is 12 cm. Find the angle which is subtended by the given arc.

A. 60; B. 90; C. 120; D. 180; E. 270;

13. What is the maximum number of points of intersection between a circle and a square that lie in the same plane?

A. 4 B. 6 C. 7 D. 8 E. 9

14. In the given figure AH is altitude and AN is angle bisector. Find x in terms of $\angle B$ and $\angle C$.



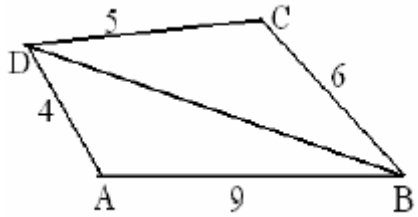
A) $|m(B)-m(C)|$ B) $|m(B)+m(C)|$ C) $\frac{|m(B)|}{2}+m(C)$

D) $\frac{|m(B)-m(C)|}{2}$ E) $\frac{|m(B)+m(C)|}{2}$

15. Of 15 players on a basketball team, one third are left handed. Out of the right-handed players there are 80% over 2 meters high. Assuming that out of the left handed players there are 40% players fewer than 2 meters. How many players are there over 2 meters in height?

a) 6
b) 7
c) 8
d) 9
e) 11

16. In the given figure, DC = 5 cm, BC = 6 cm, AD = 4 cm, AB = 9 cm what is the possible value of length BD ?



A. 4; B. 7; C. 11; D. 12; E. 13;

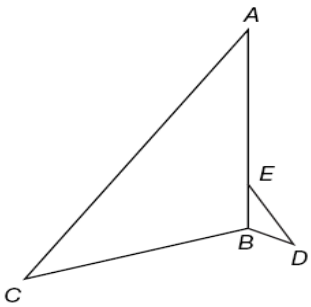
17. A parking lot has 6 spots remaining and 6 different cars in line waiting to be parked. If one of the cars is too big to fit in the two outermost spots, in how many different ways can the six cars be arranged?
(A) 36
(B) 216
(C) 480
(D) 600
(E) 720

18. What is the average (arithmetic mean) of 8 consecutive odd integers if the least of them is x?

A. x + 5; B. x + 6; C. x + 7; D. x + 8; E. x + 9;

19. Let {x} represent the fractional part of the real number x. Which of the following is definitely true?

A. $\{2x\} > \{x\}$; B. $\{x^2\} > \{x\}$; C. $\{x\} > 0.5$; D. $\{x^3\} \geq \{x\}^3$; E. $\{5x\} > 2\{x\}$;



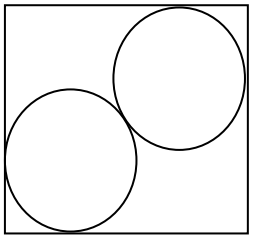
20. In the diagram above, angle *A* is congruent to angle *BED*, and angle *C* is congruent to angle *D*. If the ratio of the length of *AB* to the length of *EB* is 5:1, and the area of triangle *BED* = $5a^2 + 10$, what is area of triangle *ABC*?
A. $5a^2 + 10$
B. $25a^2 + 50$
C. $25a^2 + 100$
D. $125a^2 + 250$
E. cannot be determined

21. Which of the following definitions best fits to the graph of y = [x], where [x] represents the greatest integer less than x.

A. footsteps; B. spiral; C. mountains; D. oblique line; E. zigzag;

22. Find the derivative of y(x) if y = sin(sin(x));

A. cos(cos(x)); B. $\cos(\cos(x)) \cdot \sin(x)$; C. $\cos(\sin(x)) \cdot \sin(x)$; D. cos(2sin(x));
E. $\cos(\sin(x)) \cdot \cos(x)$;



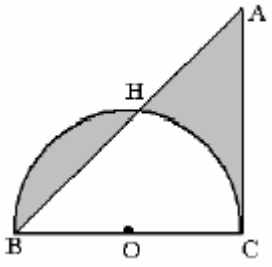
23. As the figure above shows, two identical circles are inscribed in a square. If the radius of the circle is *r*, what is the length of the diagonal of the square?

(A) $r(4+\sqrt{2})$
(B) $r(2+\sqrt{2})$
(C) $2r(1+\sqrt{2})$
(D) $4r\sqrt{2}$
(E) $5r$

24. The length, width, and height of a rectangular solid are in the ratio of 3 to 2 to 2. If the volume of the solid is *V*, what is the height, in terms of *V*?

(A) $\sqrt{\frac{V}{3}}$ (B) $\sqrt[3]{\frac{V}{3}}$ (C) $\sqrt{\frac{2V}{3}}$ (D) $\sqrt[3]{\frac{2V}{3}}$ (E) $\sqrt[3]{\frac{4V}{3}}$

25. In the given figure AC = BC = 2r and the sum of the shaded areas is 54 cm^2, then find *r*.



A. $\sqrt{3}$; B. 2; C. $\sqrt{6}$; D. $\sqrt{10}$; E. $3\sqrt{6}$;

26. Find the derivative dy/dx of the function y = $\sin^2(x^3)$.

A. $6x^2\sin(x^3)\cos(x^2)$; B. $2x^3\sin(x^3)\cos(x^3)$; C. $6x^2\sin(2x^3)\cos(x^3)$;
D. $3x^2\sin(2x^3)$; E. $6x^2\cos(2x^3)$;

27. $2\sin x\cdot\cos x\cdot\cos 2x\cdot\cos 4x\cdot\cos 8x$ =

A. cos(16x)/8; B. cos(8x)/4; C. cos(32x)/16; D. cos(8x)/16; E. sin(16x)/8;

28. The solution set of the equation $2+\cos 2x = 3\cos x$ in the interval [0,2π] is …

A) $\left\{0,\frac{\pi}{3},\frac{5\pi}{3},2\pi\right\}$ B) $\left\{\frac{\pi}{6},\frac{5\pi}{6},\frac{5\pi}{3}\right\}$

C) $\left\{0,\frac{2\pi}{3},\frac{4\pi}{3},2\pi\right\}$ D) $\left\{\frac{\pi}{2},\frac{5\pi}{4},\frac{7\pi}{6}\right\}$ E) $\left\{\frac{\pi}{4},\frac{3\pi}{4}\right\}$

29. Each minute an ameba divides into two identical. If after we put one ameba to the retort it will be full in an hour. In how many minutes it will be full if it were two amebas initially?

A. 30; B. 59; C. 61; D. 15; E. 120;

30. Which of the following is a valid conclusion?

I. All rainy days are cloudy.

II. Today is not cloudy.

A) All rainy is not cloudy.
B) Today is rainy.
C) Today is not rainy.
D) Today is cloudy.
E) None of the above is correct.